Counted but not known

Machine-only building coverage, two measurement artifacts in the transport metric, and a reproducible methodology — Bias Bounty: Mapping Equity Challenge

Zindi username **dodo25** · associated submission **pwM1ZRTe** (public 0.06391, private 0.06457) · code and data: github.com/tdogukan470-jpg/zindi-bias-bounty-dodo25 · 1 November 2026, version 4 (supersedes all earlier versions)

> **Headline.** In 1,816 tracts that are home to 7.0 million people, more than 90% of the mapped buildings are Microsoft machine-detected footprints that no person has mapped: no type, no class, no second source. The scorecard's building component counts footprints against a Microsoft reference, so it reports these tracts as complete. It cannot tell them from fully mapped ones. Machine-only coverage follows **low building density** and **tribal areas**, and it reaches **exurban as well as rural** tracts.

Executive summary

This document is submitted for both special prizes. **Part I** is the Bias Discovery and **Part II** documents the leaderboard methodology. Every number is computed from the challenge data and from our own scored submissions. No TIGER, Microsoft, HIFLD or CBP data was used anywhere. Every table and figure is regenerated by a script in the repository (Section 12).

Finding 1 — Counted but not known. Across the four scored regions (9,377 tracts with buildings, 38.8 million residents), 38% of the average tract's buildings are typeless machine-detected footprints.

- **Size of the gap.** The share is 73% in rural tracts against 30% in urban ones, and 80% in tribal statistical areas. Only 4% of buildings in the average rural tract carry any attribute, against 30% in urban tracts.
- **What drives it.** We fit regressions with region fixed effects, county-clustered standard errors and building density as a control.
  - The rural effect shrinks from 0.42 to 0.14 but stays clear (t = 4.1). Each tenfold drop in building density adds 0.16 to the machine-only share.
  - The tribal effect is 0.33 (t = 4.3), identified within Eastern Oklahoma.
  - Wildfire exposure adds nothing once rurality and density are controlled. So the exposed population is defined by density and tribal geography, not by hazard.
- **Who lives there.** 60% of the people in tracts above 90% machine-only live in tracts the Census classes as urban, mostly the exurbs of San Antonio, Houston and Austin.
- **Why the scorecard misses it.** We tested this against the hidden reference through the leaderboard.
  - The organiser's composite gap is higher in machine-only tracts, so the scorecard registers these tracts as worse. But 84–90% of that signal is region, rurality and density.
  - The scorecard's building component compares footprint counts with a Microsoft reference, and in these tracts the footprints *are* Microsoft's. So they look complete: count coverage is a median 1.01 with the narrowest spread of any group.

    - The scorecard therefore sees these places as worse for reasons unrelated to building content, and sees their buildings as fine.
- **Why it matters.** For evacuation planning, shelter identification and damage assessment, a footprint without a type is not coverage.

Finding 2 — A boundary double-count inflates rural transport gaps. Tract boundaries follow highways. A reference road on a shared boundary is counted in both tracts, while Overture's slightly offset copy is counted in one, so the neighbour shows a gap that does not exist.

- Under our reconstruction the artifact adds +0.066 to the transport component and is 0.046 larger in rural tracts. This holds with region fixed effects and clustered errors ($t = 4.8$).
- It accounts for about 29% of the rural–urban difference in our final predictions.
- One leaderboard probe supports partial double counting in the organiser's pipeline. It does not fix the magnitude.

Finding 3 — Ramps. Overture counts interchange ramps (`subclass = link`) within the counted road classes, while TIGER codes them outside the reference. This hides small gaps, more in rural tracts ($t = 7.2$ clustered). It is documented in the data and in a QGIS inspection, but it was not tested on the leaderboard.

Methodology. We had no training labels, so the method has three parts:

1. A deterministic transport reconstruction from Overture geometry.
2. Interpretable building and POI proxies.
3. A leaderboard-probe calculus whose predictions matched realised public scores to within 0.00015 on the last four re-fits.

The public score went from 0.0790 to 0.0639 (private 0.0646). The final file is rebuilt from the shipped scored vectors by one script, to within $2 \cdot 10^{-5}$. We also report two things learned after the close: where the calculus overfit the public split, and that Eastern Washington is scored on the private board only.

Caveat. The closing leaderboard shows an early file for dodo25 because of a final-selection error on our side. The method's scores are those of `pwM1ZRTe`, which can be verified on the submissions page.

Definitions used throughout

- **Organiser data**: the files on Source Cooperative (`reference/`, `strata/`). **Derived data**: our per-tract aggregates of them, shipped in the repository (`data/PROVENANCE.md`).
- **Reference proxy** (ours): Overture road segments with a route tag for an Interstate, US or State route (`core`) or a Texas FM/RM route (`fm`), excluding `subclass = link`.
- **Numerator**: Overture `motorway`, `trunk`, `primary` and `secondary` segments, all subclasses.
- **Lengths**: **L0** is the length inside the tract polygon; **L8** and **L25** are the lengths inside the polygon buffered by 8 m and 25 m. **Lbd8** is the length within 8 m of the tract boundary, counted on both sides of it.
- **Transport defined (model, Part II)**: core-route reference within the buffers > 100 m (6,269 scored tracts).
- **Transport defined (artifact analysis, §2–3)**: core + FM reference, strictly inside, > 100 m (6,707 tracts).
- **TIGER feature codes**: S1100 primary road, S1200 secondary road, S1630 ramp.
- **ML-only share, attribute share, multi-source share**: defined in §1.2.

- **Building density**: $\log_{10}$ of Overture buildings per km² of land.
- **Tribal**: `tribal_statistical > 0`; 792 of 794 such tracts are Oklahoma Tribal Statistical Areas.
- **Welch t**: two-sample t without an equal-variance assumption. **Cohen's d**: difference in means divided by the pooled SD, $\sqrt{(((n_1-1)s_1^2+(n_2-1)s_2^2)/(n_1+n_2-2))}$.
- **Clustered t**: OLS t-statistic with standard errors clustered by county (298 counties).
- **Multiple comparisons**: Part I reports 52 group comparisons. We use a Bonferroni threshold of $p < 0.05/52 \approx 10^{-3}$ and say where a result fails it.
- **Tract mean** vs **building-weighted** vs **population-weighted**: stated wherever it matters.

PART I — BIAS DISCOVERY

1. Finding 1: Counted but not known

1.1 What the scorecard measures, and what it cannot

The challenge's building component compares the number of Overture building footprints in a tract with a reference count, Microsoft US Building Footprints (per the challenge data page). The automated scorecard then reports disparity ratios of the composite score across fixed strata. Two properties of the building layer are invisible to this design:

1. **Content.** A count does not know whether a footprint is a house, a school, a fire station, a barn or a carport. Overture records this as `subtype`/`class`. Machine-detected footprints carry none of it.
2. **Provenance.** Overture's building theme merges four kinds of source: OpenStreetMap, Esri Community Maps, and machine-detected roofprints from Microsoft and Google.
    - The documentation gives OpenStreetMap "the highest conflation priority" and uses the machine sources "to fill in coverage" (Overture buildings guide, retrieved 1 Nov 2026).
    - In the bundle, 2% of Microsoft footprints are matched by a second source, against 72% of OpenStreetMap buildings.
    - The Microsoft share of a tract's buildings therefore measures how much of its building stock no OpenStreetMap contributor has mapped. Esri and Google together are 6% of the layer.
    - Where that share is high, the footprints being counted come from the same producer as the reference they are counted against.

1.2 Measures and data

All measures come from per-tract aggregates of the organiser's Overture buildings and places files (shipped as `data/derived/`), the tract table (`pop_total`, `ur_class`, `ALAND`) and the strata table. For each tract:

- **ML-only share** = buildings with `src = "Microsoft ML Buildings"` / all buildings. 100% of these footprints have a null `subtype`.
- **Attribute share** = buildings with a non-null `subtype` / all buildings.
- **Multi-source share** = `n_multisrc` / `n`, the share of buildings matched by more than one source.
- **Count coverage** = Overture buildings / `usfs_BuildingCount_sum`.

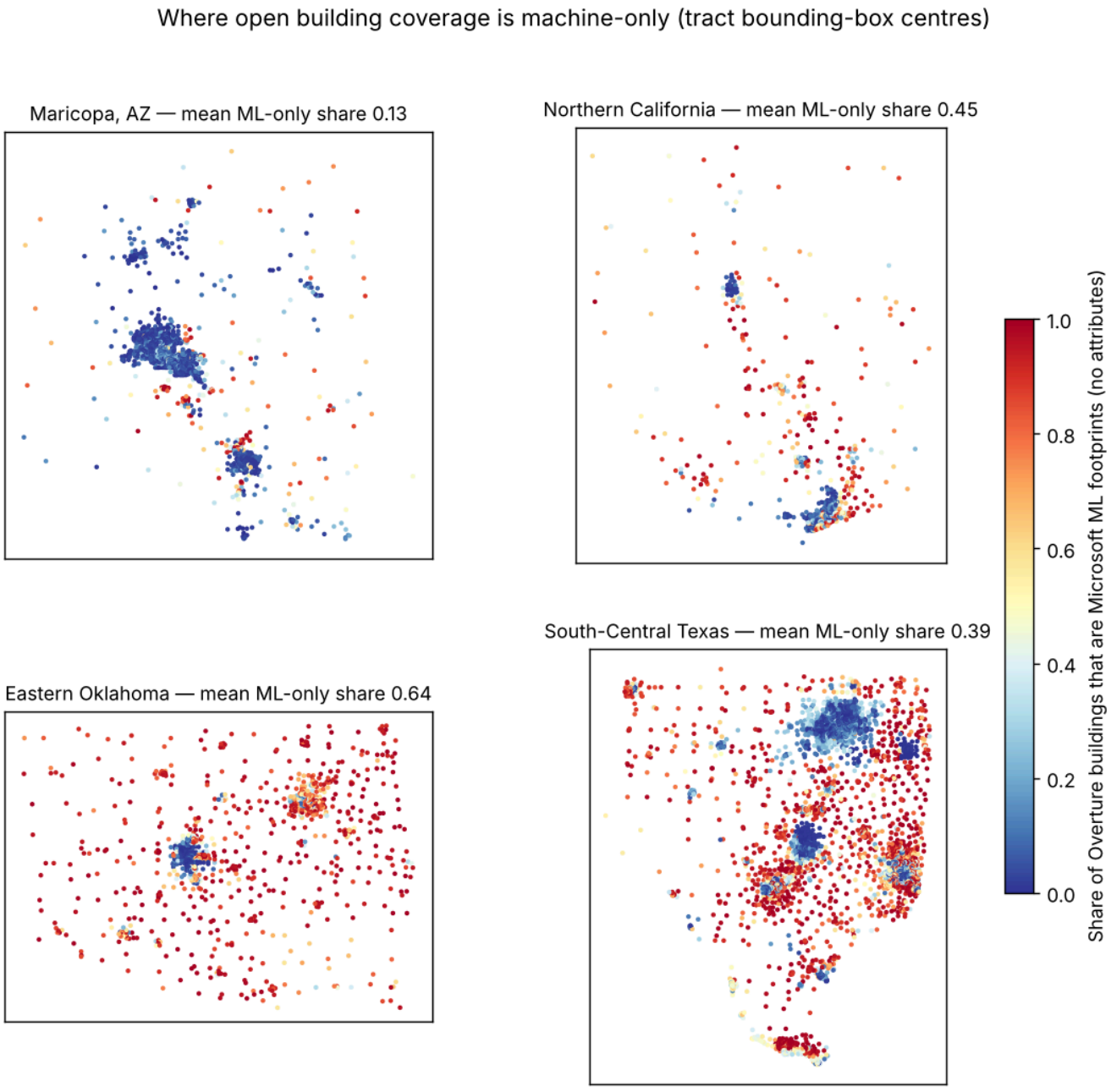
    - The USFS Wildfire Risk to Communities building count is a 30 m raster count of footprints.
    - Its second edition derives the count from the USA Structures and ONEGEO inventories (metadata retrieved 1 Nov 2026); its first edition derives it from Microsoft footprints.
    - The bundle does not say which edition it carries, so we use the count only as a second footprint inventory, not as ground truth.
- **POI measures**: share of POIs from `meta` (Facebook business pages), mean `confidence`, share with `confidence < 0.5`.

Strata are the scorecard's own: rural (`ur_class`), tribal, wildfire exposure top decile (`usfs_Exposure_mean`), ever burned (`usgs_wildfire_ever`), SVI ≥ 0.75, CVI top decile, summer heat days top decile. Shares are defined for the 9,377 tracts with at least one building.

Outputs:

- `results/quality_gap_table.csv`: 42 marginal comparisons.
- `results/regressions.csv`: conditional models.
- `results/quality_gap_numbers.json` and `results/robustness.json`: every other number.

1.3 Results: the marginal pattern



Share of Overture buildings that are machine-only footprints, by tract (bounding-box centres). Blue = community-mapped, red = machine-only.

Table 1. Building-layer content by scorecard stratum: tract means, marginal comparisons, Cohen’s d with pooled SD. Every comparison in this table passes the Bonferroni threshold before clustering. §1.4 shows which survive region, density and clustering.

Stratum	n	ML-only share, in	rest	d	Attribute share, in	rest	d	Multi-source share, in	rest	d
Rural	1,712	**0.727**	0.303	**1.22**	**0.041**	0.296	−0.80	**0.141**	0.529	−1.23
Tribal	794	**0.801**	0.341	**1.27**	0.067	0.267	−0.61	0.124	0.489	−1.09
Wildfire exposure top 10%	939	0.702	0.344	0.97	0.047	0.272	−0.69	0.154	0.492	−1.01
Ever burned (USGS)	1,211	0.664	0.338	0.89	0.072	0.276	−0.63	0.208	0.495	−0.85
CVI top 10%	938	0.520	0.365	0.41	0.069	0.270	−0.61	0.378	0.467	−0.26
Heat days top 10%	997	0.443	0.373	0.18	0.097	0.268	−0.52	0.515	0.451	+0.18
SVI ≥ 0.75	3,233	0.417	0.361	0.15	0.210	0.270	−0.18	0.440	0.468	−0.08

Population-weighted shares are almost identical, so the pattern is not an artifact of averaging over tracts:

Stratum	ML-only share, in stratum	ML-only share, rest
Rural	0.709	0.302
Tribal	0.800	0.333
Wildfire	0.690	0.334

Regional means of the ML-only share are 0.64 (Eastern Oklahoma), 0.45 (Northern California), 0.39 (South-Central Texas) and 0.13 (Maricopa).

Table 2. By region and stratum (tract means), with resident population.

Region	Stratum	Tracts	ML-only	Attribute	Multi-source	Population
Eastern Oklahoma	Tribal	792	0.80	0.067	0.12	2,627,624
Eastern Oklahoma	Rural	411	0.87	0.025	0.06	1,326,111
South-Central Texas	Rural	991	0.73	0.035	0.14	3,622,198
Northern California	Rural	147	0.73	0.047	0.16	497,511
Maricopa, AZ	Rural	163	0.37	0.115	0.32	536,056
All four regions	*Tribal AND rural*	366	0.87	0.026	0.06	1,189,189

In absolute terms (building-weighted): rural tracts contain 5.38 million Overture buildings, and 5.16 million of them (95.9%) have no type or class. Non-rural tracts contain 12.7 million buildings, of which 73% have none.

1.4 Robustness: what survives region, density and spatial clustering

The marginal comparisons of Table 1 have two weaknesses:

- Tracts are not independent, because neighbouring tracts share mappers and imagery vintages.
- The strata overlap with each other and with region and building density.

Table 3 therefore re-estimates them with region fixed effects, building density and county-clustered standard errors.

Table 3. OLS with region fixed effects and county-clustered t (n = 9,375 tracts, 298 counties). The marginal coefficient for rural is shown for reference.

Outcome	Rural, marginal	Rural	Tribal	Wildfire top 10%	SVI ≥ 0.75	Density (per $\log_{10}$)
ML-only share	+0.424 (9.8)	**+0.140 (4.1)**	**+0.327 (4.3)**	−0.003 (−0.1)	+0.075 (2.3)	**−0.159 (−7.7)**
Attribute share	−0.255 (−4.1)	**−0.106 (−3.8)**	+0.055 (1.6)	−0.023 (−1.5)	**−0.075 (−2.9)**	+0.082 (2.7)
Multi-source share	−0.388 (−6.9)	**−0.099 (−3.6)**	**−0.174 (−4.4)**	−0.011 (−0.8)	−0.042 (−1.2)	**+0.165 (9.1)**

Population-weighted versions give the same signs and similar sizes (`results/regressions.csv`). Five conclusions follow:

- **Density does most of the work.** Each tenfold drop in buildings per km² adds 0.16 to the machine-only share. Rurality keeps an independent effect of about a third of its marginal size.
- **Tribal areas are machine-only beyond what density and rurality explain:** +0.33 on ML-only share and −0.17 on multi-source share.
    - The effect is identified within Eastern Oklahoma, since 792 of the 794 tribal tracts are there.
    - Their low attribute share, by contrast, is explained by rurality and density: the conditional coefficient is +0.055, not significant.
- **Wildfire exposure adds nothing** once rurality and density are controlled. Within rural tracts, the top decile is at most 0.06 higher in Northern California, Oklahoma and Texas, and 0.09 lower in Maricopa. Wildfire-exposed communities are affected because they are rural and sparse, not because they are exposed.
- **Social vulnerability has a modest, significant conditional association.** It goes with a lower attribute share (−0.075, t = −2.9) and a higher ML-only share (+0.075, t = 2.3). The marginal d of 0.15 hides this, because high-SVI tracts are denser.
- **Tagging accounts for half of the attribute gap.**
    - Rural tracts have fewer OpenStreetMap buildings (24.5% of buildings, against 61.5% in urban tracts; building-weighted).
    - The OpenStreetMap buildings they do have are typed less often (16.4%, against 43.8%).
    - In a log decomposition, about half of the rural attribute deficit comes from each.
    - Giving rural tracts the urban OpenStreetMap share at rural tagging rates would lift their attribute share only from 0.04 to 0.10.
    - So the content gap has two layers: who mapped, and how much they recorded.

1.5 What the hidden reference says, and why the scorecard misses the pattern

The marginal test. Early in the competition we scored a probe on the leaderboard that was linear in the OpenStreetMap share of buildings, centred (u = 0.0130 × OSM share − 0.0074), in both directions. Its base was the transport-only reconstruction `t3` (public 0.07904, private 0.07911).

Probe	Change relative to t3	Public	Private
`t4`	lower the predicted gap where buildings are machine-only, raise it where they are community-mapped	0.08057	0.08065
`t4b`	the mirror image of `t4`	0.07781	0.07787
`t5`	`t4b` scaled 2.5×	0.07655	0.07658

The organiser's composite gap is higher where buildings are machine-only. Taken alone, the `t4b` direction could have lowered RMSE from 0.0790 to 0.0760, at an optimal multiplier of 4.3.

The conditional test. Rurality, region and density were not scored directly, but each can be written as a combination of the 75 scored submission vectors. Its inner product with the hidden residual then follows from the same quadratic identity used for the probes (Section 9; `code/probe_partial.py`).

- Partialling region and rurality out of the probe direction leaves **16%** of its attainable gain (RMSE 0.0790 → 0.0786; optimal multiplier 2.1).
- Partialling out rural × region and density as well leaves **10%** (multiplier 1.7).
- The sign survives in both cases.
- Caveat: the indicators are reproduced by the scored vectors only up to residuals of 0–32% of their variance, so these are estimates.

So the reference registers machine-only tracts as somewhat worse, but mostly for the reasons it registers rural, Oklahoma and low-density tracts as worse.

Why the building component cannot see content. In tracts above 90% machine-only, the footprints the building component counts are the reference producer's own footprints. Against the second footprint inventory in the strata table, count coverage is a median 1.009. Its spread narrows steadily as the machine-only share rises:

Machine-only share	< 10%	10–50%	50–90%	> 90%
Interquartile range of count coverage	0.186	0.138	0.108	0.093

This is what a count compared against near-identical footprints looks like. A tract whose buildings no one has typed scores like a tract that is fully mapped. Ingesting more machine output moves the count toward the reference and the building gap toward zero, while attribute share stays where it is.

What a composite or count can register, and what it cannot:

	Registered by the scorecard	Not registered
Machine-only tracts	Somewhat higher composite gap, mostly via region, rurality and density	That 90–99% of their buildings are untyped and unconfirmed
Building component	Count parity with the reference	Whether the counted footprints carry any content

1.6 Who is affected

- **Exposed population.** 1,816 tracts and 7.0 million people are above 90% machine-only.
  - 4.19 million of them (60%) live in the 983 tracts that `ur_class` labels urban; 721 of those tracts are in South-Central Texas (the exurban fringe of San Antonio, Austin and Houston).
  - 2.79 million live in the 833 rural tracts.
  - This is a low-density gap that crosses the urban–rural line, not a rural-only one.
- **Tribal areas.** 2.64 million residents; 2.17 million live in tracts where fewer than 10% of buildings carry any attribute. The largest Oklahoma Tribal Statistical Areas by tract count are Creek, Cherokee, Chickasaw, Choctaw, and Kiowa–Comanche–Apache–Fort Sill Apache. They form the one group with a machine-only excess beyond density (§1.4).
- **Rural residents.** 5.98 million; 5.35 million live in tracts under the same 10% threshold.
- **Wildfire-exposed residents.** 3.33 million are in the top exposure decile and 2.97 million are under the 10% threshold. The exposure follows their rurality (§1.4), but the operational stakes (§1.7) are highest for them.

The eight most populous tracts above 95% machine-only that also lie in the wildfire decile or a tribal area:

GEOID	County	Population	Buildings	…with a type	ML-only
48493000404	Wilson, TX	7,686	4,706	37	98%
48471790401	Walker, TX	7,277	3,365	11	99%
48091310903	Comal, TX	7,241	3,770	2	97%
48291700400	Liberty, TX	7,201	5,012	21	97%
48471790302	Walker, TX	7,092	4,202	18	96%
40009966200	Beckham, OK (Cheyenne and Arapaho OTSA)	7,078	5,524	40	97%
48099010604	Coryell, TX	6,971	3,039	3	99.7%
48339694002	Montgomery, TX	6,899	4,930	63	98%

1.7 Why it matters for disaster preparedness and climate resilience

The operational uses that motivate the challenge depend on attributes, not on footprints.

- **Evacuation planning.** A planner needs to know which footprints are dwellings and how many people a road serves. A tract with 4,000 typeless footprints gives a count of roofs,

not of households: barns, carports, pump houses and homes are indistinguishable. In the rural tracts of the wildfire decile, 96.5% of footprints are in this state (building-weighted).

- **Shelter and critical-facility identification.** In the building layer this relies on the `school`, `church`, `fire_station` and `hospital` classes, which machine-detected footprints never carry.
  - The places layer partly compensates: 65% of rural tracts contain a fire-department POI, against 29% of urban tracts, since rural tracts are large.
  - But rural places are more often self-reported pages: 75% of rural POIs come from `meta`, against 61% in urban tracts (d = 1.38). Their confidence is lower: 14.3% of rural POIs fall below 0.5, against 11.1%.
- **Structure triage during a wildfire, and damage assessment afterwards.** Both start from a typed building inventory. Where the inventory is typeless, they fall back to field survey, which is slowest where distances are longest.

Four places, by name (building-weighted, from the bundle):

County	Context	Overture footprints	Machine-only	No type
Butte, California	2018 Camp Fire destroyed about 18,800 structures (CAL FIRE)	105,000	75%	96%
Kerr, Texas	July 2025 Guadalupe River flood killed more than a hundred people (Texas Public Radio)	42,000	67%	84%
Gillespie, Texas	neighbours Kerr	29,000	94%	99%
Cherokee, Oklahoma	seat of the Cherokee Nation	33,000	97%	99%
Maricopa, Arizona (comparison)	Phoenix metropolitan area	1,658,000	7%	58%

1.8 Recommendations

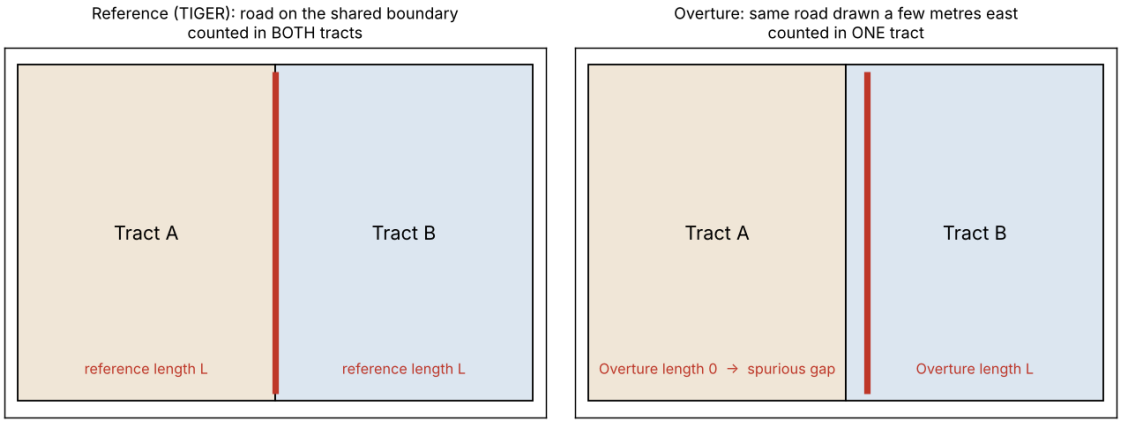
1. **Report attribute coverage and provenance alongside count coverage.** Two numbers per tract: the share of footprints with a type or class, and the community-mapped (or multi-source) share. Both are computable from the Overture extract alone.
2. **Treat the building component as not independently measurable where the candidate layer and the reference share a producer.** Flag tracts whose machine-only share exceeds a threshold, so that a low building gap there is not read as coverage.
3. **Add building density and provenance strata to the bias API.** They explain the content gap better than any current stratum except tribal area (Table 3).
4. **Pair mapping effort with tagging effort.** Raising the OpenStreetMap share alone would lift rural attribute share to about 0.10; types must be recorded too. The highest-yield targets are the tribal–rural overlap (1.2 million people, 87% machine-only) and the exurban Texas tracts above 90% machine-only.

1.9 Limitations of Finding 1

- The leaderboard scores the composite only, so we cannot apportion the reference's signal between its components. The conditional estimate in §1.5 rests on the scored vectors reproducing the region, rural and density indicators up to modest residuals.
- The tribal effect is identified only within Eastern Oklahoma.
- Attribute share depends on OpenStreetMap tagging practice (§1.4). Multi-source share does not depend on tagging, but it is not independent evidence either, since it largely re-expresses provenance (its correlation with 1 − ML-only share is 0.79).
- The edition of the USFS building count is not stated, so count coverage may itself be Microsoft against Microsoft. That would strengthen, not weaken, the argument of §1.5.
- The machine-only share is a property of the pinned Overture release (2026-08-19.0).
- Map positions are tract bounding-box centres.
- The external facts in §1.7 are context. No number in the analysis depends on them.

2. Finding 2: The boundary double-count artifact

2.1 Mechanism



Schematic of the artifact.

1. Tract boundaries frequently follow major roads. **28.9%** of our route-tagged reference length lies within 8 m of a tract boundary (two-sided; 29.7% for motorways and trunks).
2. Any intersection with closed tract polygons keeps a line that lies exactly on a shared boundary in both tracts. That includes a strict clip of road geometry to polygons. Census tract boundaries are themselves built from TIGER road edges, so reference roads coincide with them exactly.
3. Overture represents the same road a few metres off the line, so it falls inside one tract only.
4. The other tract records reference length with no matching Overture length. That is a spurious transport gap.

Test on the leaderboard.

- Our model measures the reference proxy inside 8 m / 25 m tract buffers, which reproduces double counting.
- Probe `c4` removed the buffers from an otherwise identical file. It worsened the public score from 0.06941 to 0.06971 and the private score from 0.06922 to 0.06967.
- The probe calculus puts the public optimum about one third of the way from buffered to unbuffered ($a^* = 0.35$).
- A probe with tighter buffers (`d3`) changed nothing measurable.

The reference therefore behaves as if boundary roads count on both sides, though not fully. That is consistent with exact coincidence for some roads and small offsets for others. We claim the direction for the organiser's pipeline, not the magnitude.

2.2 Size under our reconstruction, and who is affected

We computed each defined tract's gap twice: once with the reference measured inside the 8 m buffer, and once strictly inside the polygon. The difference is the artifact (n = 6,707). Under the model's own definedness flag (n = 6,269) the numbers are close: mean +0.061; rural 0.095 against 0.050.

Table 4. Boundary artifact by group.

Group	Artifact, in	rest	Welch t (iid)	Region FE, clustered: coefficient (t)
Rural	0.099	0.055	11.2	**+0.046 (4.8)**
Tribal	0.084	0.064	3.2	+0.024 (1.0), not significant
High SVI minority (≥ 0.75)	0.055	0.072	−4.9	−0.019 (−1.5), not significant

How large it is. Across defined tracts the artifact adds a mean of **+0.066** to the transport component. It exceeds 0.05 in 27% of defined tracts (19% of all tracts). Only the rural effect survives clustering.

Why it concentrates in rural tracts. Exposure does not explain it: rural tracts have *less* boundary road (34% of reference length, against 58% in urban tracts). The explanation is the clip at zero.

- In urban tracts the four-class Overture length exceeds the route-tagged reference by a median factor of 2.4, so 96% of them have a strict gap of zero. Extra reference length rarely lifts them above zero.
- Rural tracts sit at a median ratio of 1.0. There the double-counted length *creates* a gap: 63% of the rural artifact comes from tracts whose strict gap is zero (mean +0.110, against +0.050 for urban tracts at zero).
- Among tracts that already have a gap, the artifact is larger in urban tracts (0.158 against 0.085).

Effect on the composite (in our model, not the reference). The transport component has weight 1/3 in defined tracts.

- Removing the artifact from our final predictions lowers their rural/urban ratio from 2.36× to 2.20×, which removes 29% of the difference.
- The leaderboard a* of about 0.35 suggests the reference carries roughly two thirds of the buffer effect, so the share in the reference is likely smaller.
- The tribal effect is not significant once clustered, so we make no tribal claim.

2.3 Recommendation

- Assign boundary-coincident reference roads to one tract, or split them 50/50, before computing ratios.

- Report the transport disparity per stratum with and without the correction. The artifact's size depends on how close each tract sits to the clipped-at-zero region.

3. Finding 3: Ramps and link roads

Overture carries ramps and connectors as `subclass = link` within the counted classes. By length, links are 73% `motorway`, 8% `trunk`, 12% `secondary` and 7% `primary`. TIGER codes ramps S1630, outside the reference. A class-based numerator therefore credits ramp length against a reference that excludes it.

- **Size.** Under our reconstruction the effect averages +0.004 on the transport component.
- **Who it affects.** It is larger in rural tracts (+0.008; region fixed effects, clustered t = 7.2) and smaller in high-SVI-minority tracts (−0.004, t = −4.4).
    - The apparent tribal deficit in raw comparisons reverses sign under region fixed effects. It is an Oklahoma effect, not a tribal one, and we drop it.
- **Not tested on the leaderboard.** A probe without links (`c3`) was prepared but never scored.
- **QGIS inspection** (Appendix). It covered the 200 longest Maricopa `motorway`/`trunk` segments without a route tag. They are interchange ramps, collector–distributor lanes, frontage connectors and untagged second carriageways.
    - These are exactly where a class-based count and a route-based definition diverge on the ground.
    - The inspection does not measure the organiser's pipeline.

Recommendation. Exclude `subclass = link` from the numerator so that it matches TIGER's exclusion of S1630.

4. Reproducibility of Part I

Each script needs only the organiser tract, strata and ACS tables plus the repository's `data/`. They run in about a minute each with the packages in `requirements.txt`.

Script	Writes
`code/quality_gap.py <bundle> <out> [t24.csv]`	Table 1, Table 2, every other §1 number (`quality_gap_numbers.json`), the §1 figures
`code/robustness.py <bundle> data/rb3.parquet <out>`	Table 3 (`regressions.csv`), the §1.4 decompositions, count-coverage dispersion, the exposure split, the §2–3 clustered estimates (`robustness.json`)
`code/probe_partial.py <bundle> <out.json>`	The §1.5 conditional probe estimates
`code/bias_artifacts.py data/rb3.parquet <bundle> <out>`	The §2–3 tables and numbers, including the c4 probe calculus (reads t14/c4 from `data/scored_vectors.parquet`)

None of the scripts uses external data or network access, and none has a stochastic step. Four of the 200 QGIS screenshots are in the Appendix; the rest are available on request.

PART II — METHODOLOGY DOCUMENTATION

5. Data and rules compliance

- **Inputs**: the organiser's files only — tract polygons and table, strata table (227 columns), ACS housing, and the Overture roads, buildings and places extracts (release 2026-08-19.0) — plus per-tract aggregates we computed from them on Kaggle (road lengths with boundary buffers, building counts by source and type, POI summaries). File names and provenance: `data/PROVENANCE.md`; the feature kernel's log is shipped in `data/derived/`.
- **Not used**: TIGER/Line, Microsoft Building Footprints, HIFLD, County Business Patterns, or any copy of them, for scoring or for the discovery analysis.
- **Scope**: the submission template has 9,794 rows: 9,379 tracts in the four scored regions (the bundle's tract table has seven more, water-only "Census Tract 9900" polygons in Texas with no population and no buildings, which are not in the template) and 415 tracts in Eastern Washington. Part I uses the bundle's 9,386 tracts; Part II's edge-case counts (Section 11) are over the 9,379 scored ones.
- **Tools**: Python 3, pandas, numpy, DuckDB (spatial extension). Geometry ran on Kaggle CPU kernels in EPSG:5070 (CONUS Albers, equal-area, metres).
- **Determinism**: no step is stochastic; every intermediate file reproduces exactly.

6. Structure of the reference score

From the published methodology, for tract *i* with building gap B, POI gap P and transport gap T:

$$y_i = \frac{B_i + P_i + m_i\, T_i}{2 + m_i}, \qquad m_i = \mathbb{1}\left[\,\text{reference highway length in tract} > 0\,\right].$$

We treat B and P as defined in every tract. The methodology reports the share of tracts with at least one undefined component as 21% (OK), 37% (CA), 28% (TX) and 55% (AZ); our transport-undefined shares under the model's definition are 22%, 39%, 29% and 55%. The two agree within two points in every region, which leaves no room for a second undefined component of any size.

7. Transport component reconstruction

Numerator. Length of `motorway`, `trunk`, `primary` and `secondary` segments inside the tract (`ST_Length(ST_Intersection(road, tract))`, column `L0`), all subclasses.

Reference proxy. TIGER S1100/S1200 are numbered Interstate, US and State routes. We flag an Overture segment as `core` when any of its `routes[].network` values matches (exactly as in `code/routes4.py`):

```
^US:I(:.*)?$            (excluding networks containing "Future")
^US:US(:(Business|Alternate|Spur|Bypass|Truck))?$
^US:(AZ|CA|OK|TX)(:(Loop|Spur|Business|Beltway|Park))?$
^US:(OK:Turnpike|TX:Toll|TX:NTTA|TX:Harris:HCTRA)$
```

and as `fm` when it carries a Texas FM/RM network (`US:TX:FM`, `US:TX:RM`), computed in the geometry step. Ramps (`subclass = link`) are excluded from the reference proxy, because TIGER codes them S1630.

Boundary buffers. The proxy is measured inside the tract polygon buffered by 8 m (all classes) and 25 m (`motorway`, `trunk`): columns `L8`, `L25`. `Lbd8` is the length within 8 m of the boundary itself, used in Part I. The geometry step (per segment × tract, in EPSG:5070) is shipped as its output `data/rb3.parquet`; the DuckDB query is reconstructed in `code/geometry_step.sql`, since the Kaggle kernel itself was not preserved.

Gap and definedness. T = clip(1 − numerator / reference, 0, 1); m = 1 when the core-route reference within the buffers exceeds 100 m. The threshold is insensitive: 50 m or 250 m would change 13 and 39 tracts. Texas FM/RM roads enter the reference scale as separately calibrated directions (`a3`, `u2`), not through the definedness flag. The reference scale itself was calibrated per region on the leaderboard (`e1`, `e2`, `f1–f3`).

Overture tagging gap. Rescuing untagged mainline carriageways identified in the QGIS inspection changes only 94 tracts (about 1,000 km against 52,000 km of tagged core).

8. Building and POI proxies, and the final predictor

- **Building directions**: Overture counts by source (OpenStreetMap share, Microsoft share); counts without small structures (< 40 m²) against `usfs_BuildingCount_sum`; area per structure; multi-source share.
- **POI directions**: places per capita and per housing unit; source shares and confidence; facility categories (fire, EMS, schools; hospitals excluded, as the methodology states); business-type places relative to housing units; county-level POI per capita.
- **Neighbourhood**: 8-nearest-tract means of the prediction within each region (tract bounding-box centres, longitude scaled by cos latitude), applied to transport-undefined tracts.
- **Region and definedness constants, region stretch, rural indicators**: the late orthogonal probes (`rc*`, `ru*`, `st*`).

The final predictor is a linear combination of scored submissions: **$t24 = t23 + \Sigma_k a_k (s_k - t23)$**, clipped to [0, 1], over 75 directions (the 78 named scored vectors minus `t24` itself, the base `t23`, and `f4`, which changed only Eastern WA). The coefficients a_k, with each direction's noise estimate σ_k and the quantities q_k and g_k of Section 9, are in `results/t24_coefficients.csv`; the scored vectors are shipped in `data/scored_vectors.parquet` with their public and private scores in `data/scores.csv`; `python code/make_t24.py <out.csv> [<coefficients.csv>]` rebuilds t24 from them (maximum absolute difference from the submitted file $1.8 \cdot 10^{-5}$, which comes from scores rounded to six digits for the earliest files at the time).

9. Leaderboard-probe calculus

Let *s* be a scored submission with public MSE M_0 and *u* a perturbation vector over the publicly scored tracts. Then

$$\mathrm{MSE}\,(\,s + a u\,) = M_0 + 2a\,c + a^2 q, \qquad q = \bar{u^2}, \quad c = u\,(\,\bar{s} - y\,),$$

with bars denoting means over the scored tracts. *q* is computed locally; one scored probe at a = 1 gives $c = (M_1 - M_0 - q)/2$ and the optimal step $a^* = -c/q$. For many directions we solve $Qa = -c$ with the local Gram matrix $Q = (\langle u_j, u_k \rangle)$; information from earlier probes carries over to a new base exactly through $c_{new} = c_{old} + \langle u, \Delta base \rangle$ (mean inner product).

9.1 Global re-fit with a noise model

Every scored file is a vector with one known inner product against the hidden residual. We re-fit the best combination of all scored vectors at once:

$$a^{*} = -\left(G + \lambda \operatorname{diag}\left(\sigma_k^2\right)\right)^{-1} g,$$

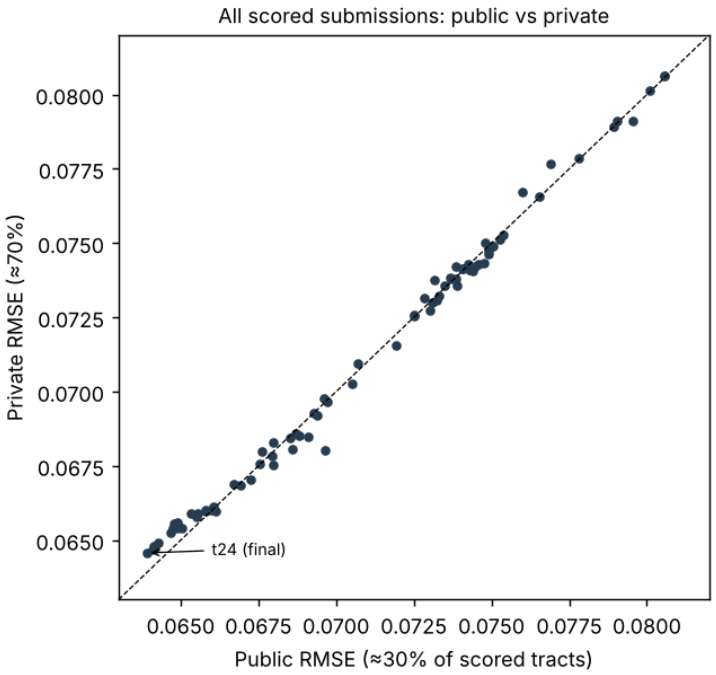
where D is the matrix whose columns are $d_k = s_k - s_base$, $G = D^{T}D / n$ is its Gram matrix over the scored tracts, $g_k = (M_k - M_base - q_k)/2$ is the inner product of d_k with the hidden residual implied by the quadratic identity, σ_k is estimated by leave-one-out prediction of each scored submission from all the others (a vector whose score the others cannot predict is treated as noisier; $\sigma_k = 2 \cdot 0.066 \cdot \max(\text{LOO error}, 2 \cdot 10^{-5})$), and $\lambda = 3 \cdot 10^{5}$ was chosen by simulation so that the *expected realised* gain under noisy g is maximised rather than the in-sample gain. The late orthogonal probes (`rc*`, `ru*`, `st*`) were given their measured, smaller noise ($3 \cdot 10^{-5}$).

9.2 Quality control that changed results

- **Leave-one-out consistency revealed two mislabeled batches** (scores attached to the wrong files: b1/b2/b4, and a1/a2). With the corrected mapping the joint prediction error for `t14` fell from 0.00102 to 0.00021, and the re-fit `t18` was predicted at 0.06601 and scored 0.06599.
- **Prediction accuracy of the calculus on the last four re-fits:** `t21` predicted 0.06486 → scored 0.06490; `t22` 0.06464 → 0.06476; `t23` 0.06408 → 0.06415; `t24` 0.06390 → 0.06391.
- **Eastern Washington.** A probe that changed only Eastern WA tracts (`f4`, 128 tracts, RMS change 0.105) left the public score unchanged to nine digits, so all public inner products are computed on the four regions. After the close its *private* score differed from its base (0.06831 versus 0.06755): **Eastern WA is excluded from the public board but included in the private one.** Our WA predictions were never calibrated; the re-fits moved them by an RMS of 0.012 between `t18` and `t24`. We state this because it bounds what Section 9.3 can conclude.
- **Public/private.** The public board is a fixed subset of the four regions; the private board adds the remaining tracts and Eastern WA. We shrink directions in proportion to their estimated noise; Section 9.3 audits how well that worked.

9.3 Where the calculus overfit the public split

With the private scores visible after the close, the method can be audited. Across all 85 scored submissions the public and private RMSE correlate at 0.9986 (median |difference| 0.00023). The structural improvements (`t3` → `t18`) transferred almost one-for-one: `t18` scored 0.06599 public and 0.06600 private. The probe-driven re-fits after it did not transfer fully: the private–public difference grows from +0.00001 (`t18`) to +0.00019 (`t19`), +0.00025 (`t20`), +0.00050 (`t21`) and +0.00066 (`t22`–`t24`). Of the 0.0021 public gain from `t18` to `t24`, about 0.0014 appeared on the private board. Two effects are confounded in the remaining 0.0007: fitting to the public sample, and the uncalibrated Eastern WA predictions that the private board includes; the `t16`/`f4` pair shows WA alone can move a private score by 0.0008. The noise model anticipated the direction of the first effect but not its size. `t24` is still the best private submission of the 85 (0.06457), so the overfit reduced the gain rather than reversing it.



Public against private RMSE for all 85 scored submissions; the dashed line is equality.

10. Results (RMSE on `coverage_gap_score`)

Submission	Content	Public	Private
`t3`	transport reconstruction with 8/25 m boundary buffers	0.07904	0.07911
`t5`	+ OSM-share direction (Section 1.5)	0.07655	0.07658
`t7–t9b`	+ source-mix and per-capita POI/building directions	0.07430	0.07411
`t10–t13`	+ structural building (USFS count), TX FM roads, denominator weighting	0.07251	0.07257
`t14`	+ primary/secondary routed roads in reference (largest single direction)	0.06941	0.06922
`t15–t17`	+ buffer and reference-scale calibration by region	0.06792	0.06782
`t18`	global re-fit, corrected labels	0.06599	0.06600
`t19`	+ county/kNN/tail/facility probes	0.06581	0.06601
`t20`	noise-weighted re-fit over all scored vectors	0.06552	0.06577
`t21–t22`	re-fits including kNN probes	0.06476	0.06542
`t23`	+ region × definedness constant probes	0.06415	0.06481
`t24 (pwM1ZRTe)`	+ rural and stretch probes	**0.06391**	**0.06457**

Directions that did not help, reported for completeness: Texas FM roads added to the reference alone; rescaling building/POI corrections by 1/(2+m) (the opposite sign was optimal); population-based facility-absence indicators (optimal weight ≈ 0); county-level POI-per-capita gap (negative weight); kNN smoothing applied to transport-defined tracts (optimal weight ≈ 0, probe `k6`); scaling Eastern WA (`f4`, no public effect, worse private).

Leaderboard caveat. The two files marked as final on the submissions page were set on 11 October and never updated, so the closing leaderboard shows dodo25 at 0.0795 / 0.0791. The method's final file is `pwM1ZRTe` (0.06391 public, 0.06457 private), which the judges can verify on the submissions page. We state this plainly rather than let the writeup disagree with the board.

11. Edge cases and alternative weightings

The prize text asks how zero-population, water-dominated and data-sparse tracts were handled. The table gives their counts among the 9,379 scored tracts of the four regions (the seven water-only tracts absent from the template are excluded) and the mean prediction of `t24` for each (overall mean 0.059).

Edge case	Definition	Scored tracts	Mean prediction	Handling
Zero population	`pop_total = 0`	26	0.145	Per-capita ratios are undefined. In code a zero or missing denominator is clipped to 1 and an undefined ratio takes its neutral value (gap contribution 0), so these tracts are predicted from the transport reconstruction, the building count against the USFS count, and the 8-NN neighbourhood mean. They receive the highest mean prediction; we regard this as a known weakness, since the reference itself is least meaningful where nobody lives.
Near-zero population	`pop_total < 100`	65	0.093	Same clipping. Small denominators are the main reason these predictions sit above average; accepted because the tracts are few and contribute little to RMSE.
Zero housing units	ACS `housing_units = 0`	80	0.098	Housing-unit ratios take their neutral value; building proxies use the USFS count instead.
Water-dominated	land area < 50% of tract area (`ALAND` / `area_m2_calc`)	43	0.114	A land-area density direction (g2) and a water-share direction (g4) were

Edge case	Definition	Scored tracts	Mean prediction	Handling
				prepared but never scored, so these tracts receive no special treatment beyond the neighbourhood mean. Listed as an open item. The seven water-only tracts are not in the template.
No Overture buildings	`n = 0`	2	0.097	Building ratios undefined; prediction from transport and neighbourhood.
Data-sparse	< 20 buildings / no POIs	5 / 2	0.105 / 0.097	Ratios computed with clipped denominators; the neighbourhood term supplies most of the signal.
USFS count = 0	`usfs_BuildingCount_sum = 0`	2	0.097	Building direction contributes zero.
Transport undefined	core-route reference within buffers < 100 m	3,110	0.023	Score structure (B + P)/2; 8-NN smoothing within region with optimal weight 0.68 on this subset (probe k4).
Bounds	—	all	—	Predictions clipped to [0, 1]. Early files used a 0.01 floor (1,450 tracts). Probe d4 set those tracts to 0 and received an optimal weight of −0.68: the floored tracts were under-predicted, not over-predicted, and later re-fits moved them up. 124 tracts of the final file sit at

Edge case	Definition	Scored tracts	Mean prediction	Handling
				exactly 0 through clipping of the combined fit.
Eastern WA	415 tracts	—	—	Predicted with the same pipeline but never calibrated: excluded from the public board, present on the private board (Section 9.2).

11.1 Assumptions, stated

1. Only `coverage_gap_score` is scored: probes that changed only the component columns left the score unchanged.
2. The score has the structure of Section 6, with building and POI components defined in essentially every tract (argument in Section 6).
3. Transport is undefined where the core-route reference is below 100 m; insensitive to the threshold (Section 7).
4. The public board is a fixed subset of the four regions' tracts, so inner products estimated on it are unbiased for those regions up to sampling noise; the private board adds the remaining tracts and Eastern WA (Section 9.2). Section 9.3 reports how far the transfer held.
5. Predictions are bounded in [0, 1], as the reference is a ratio-based score.
6. The reference is deterministic, so the quadratic identity of Section 9 holds exactly on the public sample.

Alternative weightings tested. Weighting building/POI directions by each tract's actual 1/(2+m) share versus a uniform weight: uniform was better, which suggests the organiser's B and P vary more in transport-defined tracts. Region-specific reference scales (`e1`, `e2`, `f1`–`f3`) were kept; a county-level POI direction was rejected.

12. Code map and how to reproduce

Repository: `github.com/tdogukan470-jpg/zindi-bias-bounty-dodo25` (public). Python 3.10+, `pip install -r requirements.txt`.

Path	Purpose
`data/rb3.parquet`	Output of the geometry step: per (tract, class, route flags, subclass) lengths L0/L8/L25/Lbd8 in EPSG:5070
`data/roadfeat2.parquet`	Per-tract road features derived from `rb3` (`coreB` is the model's definedness reference)
`data/scored_vectors.parquet`, `data/scores.csv`	All 78 named scored submission vectors with Zindi ids and public/private scores (85 rows; 7 early or duplicate uploads have no named vector)
`data/derived/`	Per-tract building aggregates and POI summary used by Part I; feature-kernel log; `data/PROVENANCE.md` lists organiser vs derived files
`code/geometry_step.sql`	Reconstructed DuckDB query of the geometry step (the Kaggle kernel was not preserved; its output is `rb3.parquet`)
`code/routes4.py`	Route-tag regex over `routes[].network` (historical; reads the per-segment table built on Kaggle)
`code/feat.py`, `code/cands.py`	Historical feature and direction scripts (read intermediate files from the Kaggle step)
`code/probe_solve.py`	Quadratic probe solver, ridge joint fit
`code/make_t24.py <out.csv> [coef.csv]`	Rebuilds the final submission from the scored vectors and scores, and writes its 75 coefficients
`code/robustness.py`, `code/probe_partial.py`	Part I robustness (Table 3, clustering, decompositions) and the conditional probe estimate
`code/quality_gap.py`	Part I, Finding 1: tables, numbers and figures
`code/bias_artifacts.py`	Part I, Findings 2–3: tables and numbers
`code/leaderboard_audit.py`	Section 9.3: `results/all_submissions_public_private.csv`, `figs/fig_public_private.png`
`results/`	`quality_gap_table.csv`, `quality_gap_regions.csv`, `quality_gap_numbers.json`, `regressions.csv`, `robustness.json`, `probe_partial.json`, `bias_artifacts_table.csv`, `bias_artifacts_numbers.json`, `t24_coefficients.csv`, `all_submissions_public_private.csv`
`submissions/`	Scored files t18–t24

Order: `geometry_step.sql` and the feature kernel (Kaggle) → `rb3.parquet`, `data/derived/` → model directions → scored submissions → `make_t24.py`. Part I scripts need only the organiser tract, strata and ACS tables plus the repository's `data/`.

References

- Zindi, *Bias Bounty Mapping Equity Challenge* — data page and special-prize form text. https://zindi.africa/competitions/bias-bounty-mapping-equity-challenge/data (retrieved 31 Oct 2026).
- Source Cooperative, challenge data repository. https://source.coop/humane-intelligence/bias-bounty-mapping-equity-challenge (retrieved 31 Oct 2026).
- Overture Maps Foundation, *Buildings* guide (data sources and conflation priority). https://docs.overturemaps.org/guides/buildings (retrieved 1 Nov 2026).

- U.S. Census Bureau, *MAF/TIGER Feature Class Code Definitions* (S1100, S1200, S1630). https://www.census.gov/library/reference/code-lists/mt-feature-class-codes.html (retrieved 1 Nov 2026).
- Jaffe, Scott, Callahan, Dillon, Karau, Lazarz (2024), *Wildfire Risk to Communities: Spatial datasets of wildfire risk for populated areas in the United States*, 2nd ed., USFS Research Data Archive, doi:10.2737/RDS-2020-0060-2 (metadata retrieved 1 Nov 2026).
- CAL FIRE, Camp Fire incident page (structures destroyed). https://www.fire.ca.gov/incidents/2018/11/8/camp-fire (retrieved 1 Nov 2026).
- Texas Public Radio, “More than 100 people died in Texas’ historic floods”, 8 Jul 2025. https://www.tpr.org/news/2025-07-08/more-than-100-people-died-in-texas-historic-floods-here-are-some-of-their-stories (retrieved 1 Nov 2026).

Appendix — QGIS field inspection (Maricopa)

The 200 longest `motorway`/`trunk` segments without a route tag were each rendered over the OpenStreetMap basemap with the Overture road layer styled by class (basemap © OpenStreetMap contributors). Four representative cases follow.

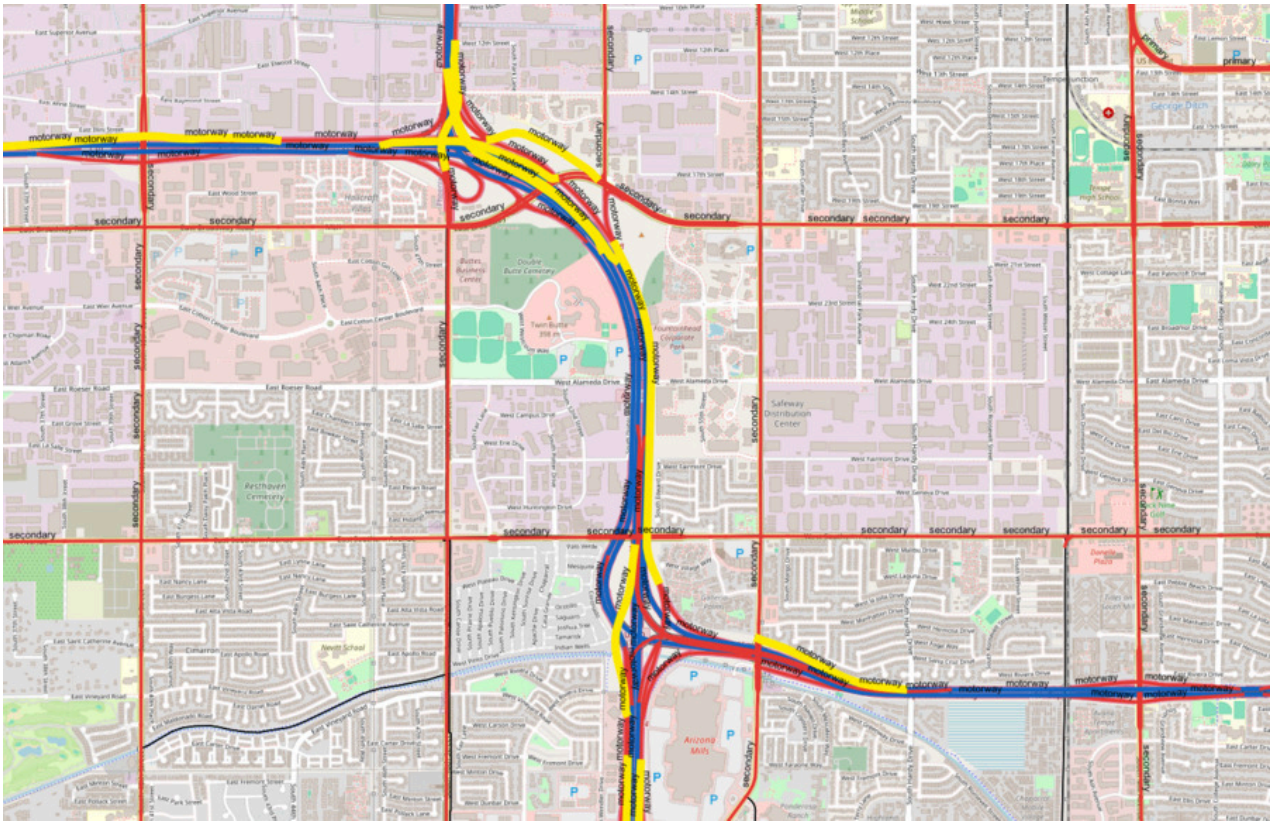


Fig. A1 — Freeway-to-freeway interchange (segment 6348, 1.8 km). Ramps and collector lanes branch from the mainline; Overture classes them `motorway`, TIGER codes them S1630.

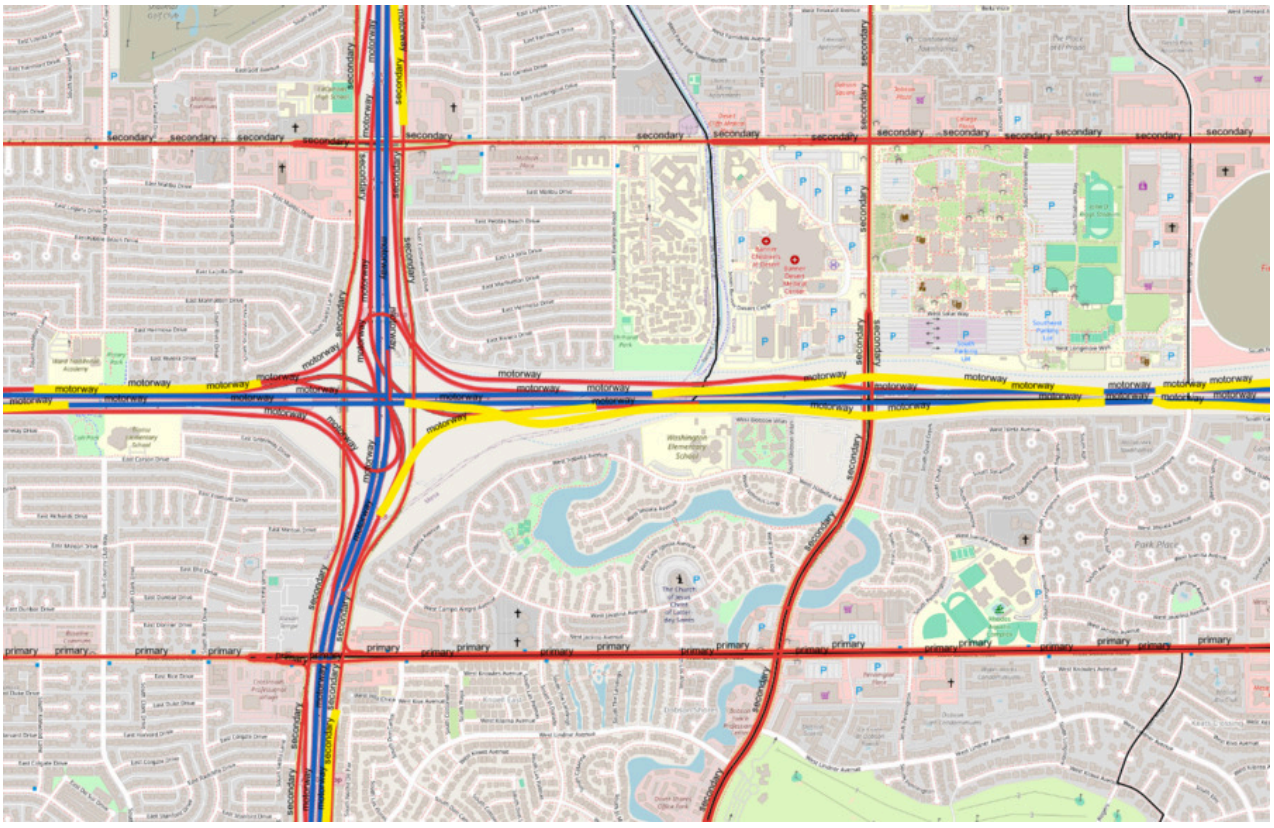


Fig. A2 — Interchange on an arterial grid (segment 9798, 1.4 km). Ramp loops and frontage/service connectors run parallel to the mainline.

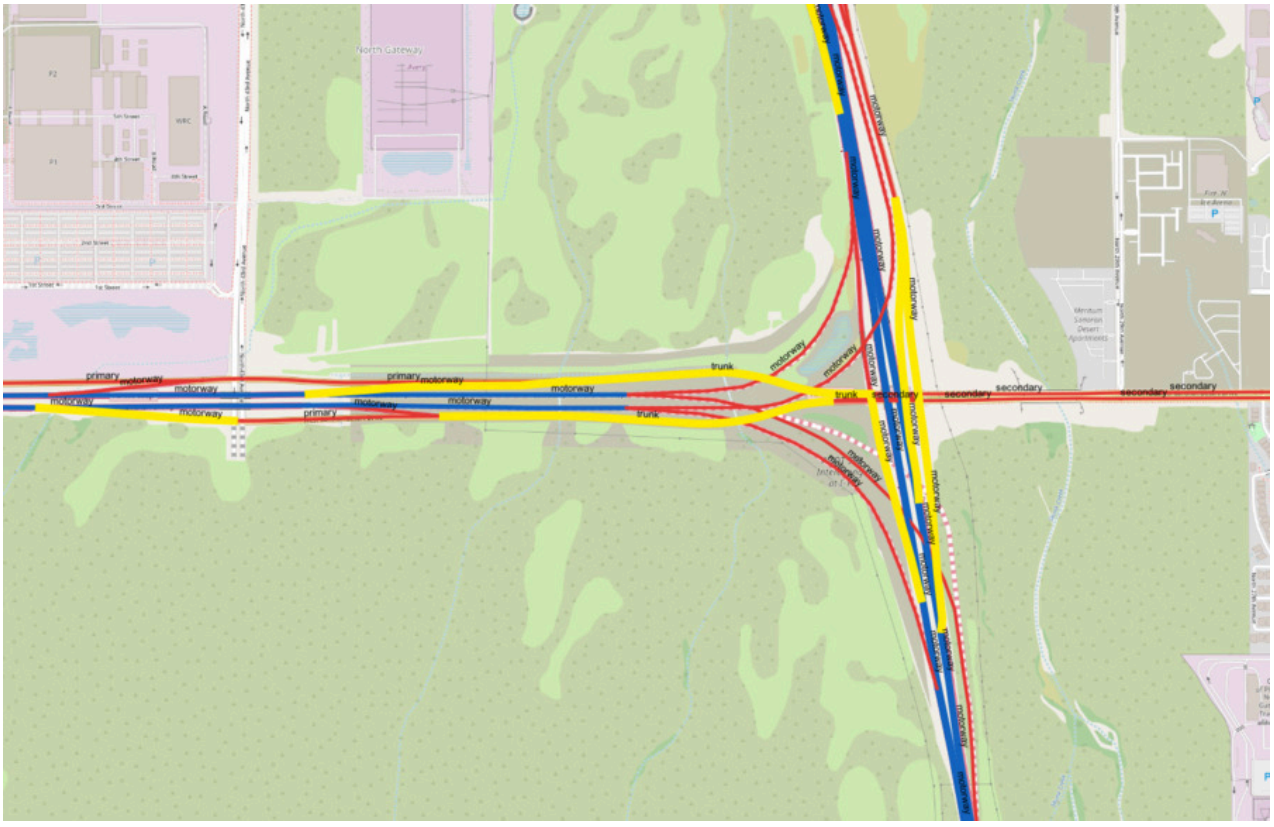


Fig. A3 — Rural trunk junction (segment 30157, 1.2 km). Link roads make up a large share of named-highway length in a sparse tract.

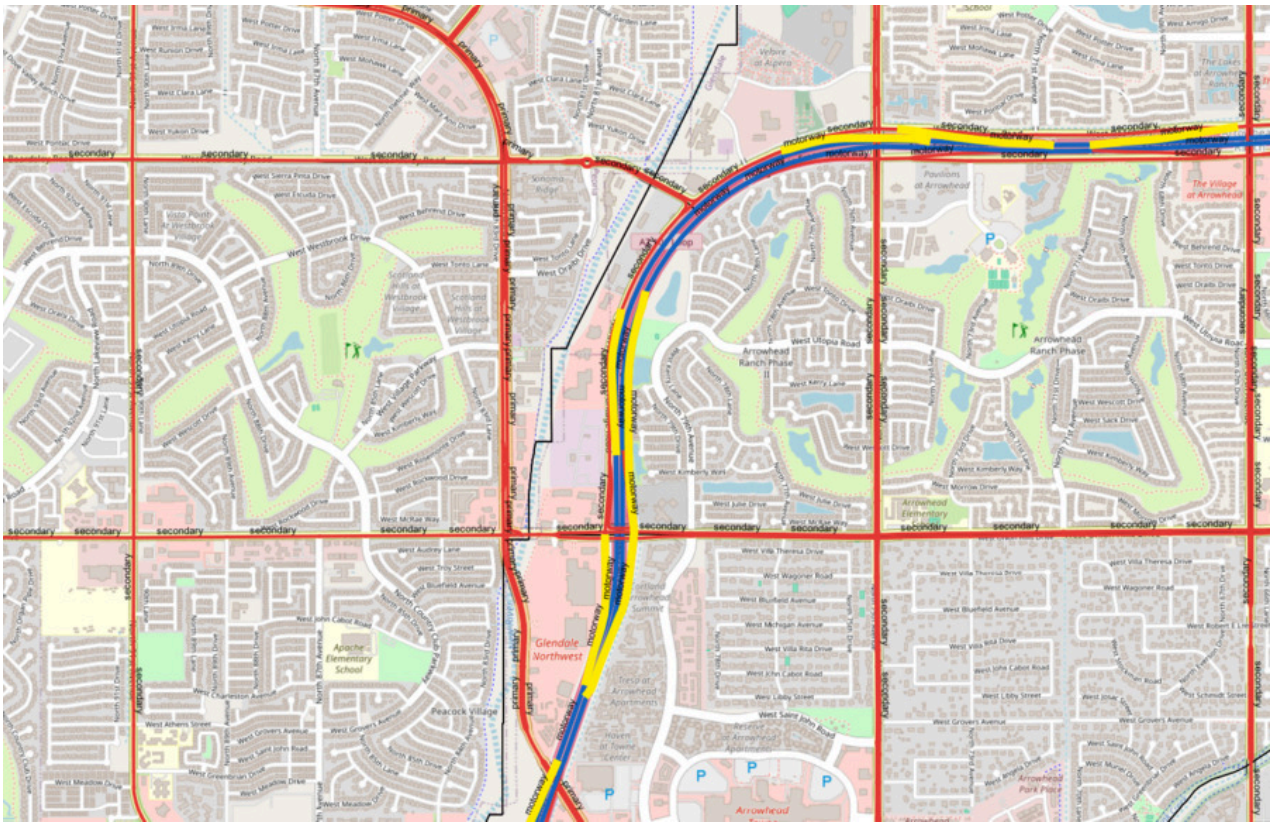


Fig. A4 — Curving divided freeway (segment 28771, 1.0 km). The second carriageway carries no route tag, so a route-based reference proxy misses it.